Example 1

Use a stem and leaf plot to show the following ages of 21 employees in a factory:
25 35 28 47 52 33 50 28 33 35 48 55 29 50 39 41 32 29 56 26 35

(i) What is the mode of the data?
(ii) What is the median age?
(iii) What is the range?

Since the youngest is 25 and the eldest is 56, we will use the stems 2, 3, 4, 5

Un-Ordered

Stem	Leaf
2	5 8 8 9 9 6
3	5 3 3 5 9 2 5
4	7 8 1
5	2 0 5 0 6

Ordered

Stem	Leaf
2	5 6 8 8 9 9
3	2 3 3 5 5 5 9
4	1 7 8
5	0 0 2 5 6

Key 2|8 = 28

(i) The mode is 35 as it occurs most frequently
(ii) The median age (of 21) is the 11th age in increasing order.
This value is 35, i.e., the median is 35 years.
(iii) The range of this data is 56 − 25 = 31 years

Exercise 14C

1. The number of points scored per match by Dragons basketball team are shown below:

85	67	56	69	99	97	59	65	84	97
49	72	89	78	66	81	92	88	53	73

4	9
5	3 6
6	
7	
8	
9	

(i) Copy and complete the (ordered) stem and leaf plot on the right.
(ii) How many matches did they play?
(iii) Find the range of scores.
(iv) In what percentage of matches did they score more than 80 points?

2. Twenty four pupils were asked how many computer games they had in their collection. The results are shown below:

23	2	18	14	7	4	25	21	32	26	31	6
17	6	18	19	31	21	12	1	0	8	14	15

(i) Draw a stem and leaf diagram to represent this information.
(ii) How many pupils had more than 20 games?
(iii) How many students had fewer than 5?
(iv) Is there a mode for this data? Explain your answer.
(v) Calculate a mean for this data correct to the nearest whole number.

3. The marks of 24 students in a test were:

41	23	35	15	40	39	47	29
52	54	45	27	28	36	48	51
59	65	42	32	46	53	66	38

Copy and complete the stem-and-leaf diagram.

Find: (i) the range of the marks
(ii) the median score of 24 students. This is obtained by adding the 12th and 13th score and dividing by 2 i.e., $\left(\frac{12th\ score + 13th\ score}{2}\right)$

Stem	Leaf
1	
2	3
3	
4	1
5	
6	

Key 4|1 means 41

4. A stem and leaf plot of the masses, in kg, of people in a waiting room is given.
(i) How many people were in the waiting room?
(ii) Write down the range of their masses.
(iii) Find the median mass.

Stem (tens)	Leaf (units)
3	2 5
4	1 1 3 7 8
5	0 2 5 8
6	4 8
7	1
8	2

Key 7|1 = 71 kg

5. The results of a science test are shown in this (unordered) stem and leaf plot.
(i) Rewrite this diagram putting the leaves in order.
(ii) Find the range of the results.
(iii) Find the median result using $\left(\frac{10th\ result + 11th\ result}{2}\right)$ of the new plot.

Stem	Leaf
2	4 7 8 5
3	0 2
4	2 7 3 3 6 9 5
5	4 5 4 0
6	1 2
7	3

Key 4|2 = 42

6. During a swimming class a test was done to find out how long a student could hold their breath.
The swimming instructor presented her results as a stem and leaf plot.
(i) How many students were in the class?
(ii) How many students held their breath for 45 or longer?
(iii) How many students held their breath for less than 20 seconds?
(iv) What was the range of times for holding your breath?
(v) Show that the median time for holding your breath was 38 seconds.

Stem	Leaf
0	6
1	2 3 4 7
2	1 1 3 6
3	2 4 6 7 7 9
4	0 0 2 3 3 5 9
5	1 2 5
6	2 4
7	4

Key 3|2 = 32 s

7. Represent the following data as an ordered, stem and leaf plot:
24, 30, 46, 33, 5, 17, 45, 39, 41, 8, 28, 12, 24, 25, 15
From your plot find the: (i) range (ii) median (iii) mode of this data.

8. The following is a list of the results of a maths test for a group of students:

63	58	63	52	59	65	69	75	70	54
57	63	76	81	63	68	59	40	65	74
80	44	47	53	70	81	68	49	57	61

(i) Andre made a stem and leaf plot of these results as shown over. Which two results did he leave out of the plot?
(ii) Construct a complete (ordered) stem and leaf plot of these results.
(iii) Find the (a) the mode and (b) the median of the data
(iv) What is the range of the data?

4	0 4 7 9
5	8 2 9 4 7 9 3 7
6	3 3 5 9 3 3 8 5
7	5 0 6 4 0
8	1 0 1

Key 8|1 = 81